

polyroots

Operating Envelope & Limitations Report

Phase-1 CPU reference implementation — argument-principle subdivision root finder

Empirically measured June 16, 2026 • build: Release / -O2 / C++17

1. Executive Summary

The solver locates polynomial roots by recursively subdividing a bounding square and counting enclosed roots via the argument principle (winding number), then polishing with modified Newton. Within a well-defined envelope it is accurate to machine precision. This report maps that envelope: along five axes — degree, root magnitude, coefficient dynamic range, root clustering, and multiplicity — it records the exact point at which the algorithm degrades or fails, and ties each cliff to its cause in the code.

The single most restrictive limit is coefficient dynamic range: roots spanning more than ~2–3 orders of magnitude are silently lost. The most *dangerous* limit is high multiplicity (> 8), which can emit phantom roots or hang rather than fail cleanly.

Safe Operating Envelope		
Degree (well-conditioned)	$\leq 100\text{--}150$	runtime $\approx O(n^{2.7})$; ~6 s at $n=100$
Degree (ill-conditioned)	$\leq 15\text{--}20$	Wilkinson already fails at degree 20
Root magnitude	$ \text{root} \leq 10^{(300/n)}$	coefficient overflow in construction
Dynamic range (binding)	$\leq 10^2\text{--}10^3$	Horner cancellation near small roots
Multiplicity per root	≤ 8	noise-floor heuristic misfires beyond this
Cluster separation	any (count) / $\geq 10^{-6}$ (resolved)	merges but never loses/invents

2. Methodology

Each axis was swept with a dedicated probe until the solver broke. Correctness was judged against construction truth (polynomials built from known roots via `from_roots`) and closed-form roots of unity; no external solver was used, since a double-precision reference (e.g. a companion-matrix eigensolver) shares this solver's cancellation failure modes and so is not independent on the hard cases. Runtime was wall-clock, Release build, single thread.

3. Axis-by-Axis Findings

3.1 Degree — bounded by runtime, not accuracy

For well-conditioned roots (roots of unity), accuracy holds at machine precision at every degree tested. What taps out is **speed**: cost per box is $\sim O(n^2)$ (about $4n$ boundary samples $\times O(n)$ Horner each) and the box count grows with n , giving an overall $\sim O(n^{2.7})$ trend.

Degree n	Roots found	Time	Max error
5	5 / 5	0.005 s	2e-16
20	20 / 20	0.078 s	6e-16
40	40 / 40	0.43 s	6e-16
60	60 / 60	1.37 s	8e-16
80	80 / 80	3.09 s	7e-16
100	100 / 100	5.87 s	9e-16

Caveat: this is the well-conditioned ceiling. For ill-conditioned polynomials the correctness limit is far lower — the Wilkinson polynomial fails at degree 20 due to catastrophic cancellation (a documented Phase-1 gap).

3.2 Root magnitude — hard cliff at $10^{(308/n)}$

Three real roots at radius 10^k were used. Accuracy is perfect up to 10^{100} , then fails completely at 10^{150} . Cause: from_roots forms the constant term $c_0 = \prod \text{roots}$; for a cubic this is $\sim 10^{3k}$, which overflows a double (1.8×10^{308}) near $k \approx 102$. The bound becomes infinite and the solver gives up. The threshold scales inversely with degree: $|\text{root}| \leq \sim 10^{(308/n)}$.

Roots ~ 10^k	Roots found	root_bound	Time
10^{64}	3 / 3	1.2e65	0.000 s
10^{100}	3 / 3	1.2e101	0.000 s
10^{150}	0 / 3	inf (overflow)	0.000 s
10^{200}	0 / 3	inf (overflow)	0.000 s

3.3 Dynamic range — the weakest axis (breaks at $\sim 10^3$)

Roots placed at $\{10^{-k}, 1, 10^k\}$. Only ~2–3 orders of magnitude of spread are tolerated before roots are **silently lost**. Near the small roots, Horner's partial sums reach $\sim |c_1 \cdot z|$ while $|P|$ itself is ~ 1 , so $\arg(P)$ is dominated by rounding noise; those boxes register as empty and are pruned. This is the binding real-world constraint because innocuous-looking polynomials hit it.

Spread (max/min)	Roots found	Outcome
10^2	3 / 3	all recovered
10^4	2 / 3	1 root lost
10^8	1 / 3	2 roots lost
10^{16}	0 / 3	all lost

3.4 Clustering — count robust, resolution stops at $\sim 10^{-6}$

Two roots separated by distance d , plus two well-separated roots. The total multiplicity is **always correct**, even at $d = 10^{-12}$ — clustered roots are never lost or invented. Below $\sim 10^{-6}$ – 10^{-8} the solver simply stops

separating them, reporting one low-confidence box of multiplicity 2. This is graceful degradation, not a failure.

Separation d	Total found	Resolved as distinct?
10^-4	4 / 4	yes (2 points)
10^-6	4 / 4	yes (2 points)
10^-8	4 / 4	no (merged to 1)
10^-12	4 / 4	no (merged to 1)

3.5 Multiplicity — reliable to m=8, then degrades, then explodes

For $(z-1)^m$, two limits stack. (a) **Position accuracy** is capped at $\sim \epsilon^{(1/m)}$: even at $\epsilon=10^{-10}$, $m=8$ yields only ~ 2 correct digits. (b) **Count integrity** holds to $m \approx 8$, then the noise-floor heuristic misfires — first undercounting, then spawning **phantom roots** ($m=24$ reports 42), and finally **hanging** ($m=32$ did not terminate). This is the only axis that fails un-gracefully.

Multiplicity m	Count found	Position error	Status
4	4 (correct)	2.5e-4	ok
8	8 (correct)	2.1e-2	accuracy ceiling
12	8 (lost 4)	—	undercount
16	6 (lost 10)	—	undercount
24	42 (phantoms)	—	spurious roots
32	—	—	hangs / no termination

4. Root-Cause Cross-Reference

Symptom	Mechanism	Locus in code
Slow at high degree	$O(n)$ samples/side $\times$ $O(n)$ Horner per box	winding.cpp (sample count)
Roots lost above 10^{100}	constant-term product overflows double	polynomial.cpp (from_roots)
Wide-range roots lost	cancellation corrupts $\arg(P)$, box pruned	winding.cpp (noise floor)
Clusters not separated	subdivision halts at noise floor	solver.cpp (trusted flag)
Phantom roots / hang at high m	noise region subdivides without bound	solver.cpp (no work-list cap)

5. Recommendations

• **Variable scaling (balancing).** Substitute $z = \alpha w$ with $\alpha \approx \text{root_bound}$ so roots land near $|w| \leq 1$. This shrinks the contour (fewer subdivision levels $\rightarrow$ faster), and bounds Horner partials near 1^n instead of R^n — directly attacking both the runtime and overflow limits.

- **Compensated / scaled Horner (Phase-2).** Error-free transformation evaluation would raise the noise floor, recovering the wide-dynamic-range and Wilkinson cases — the dominant real-world limitation.
- **Work-list / depth cap in solver.** A hard iteration or box-count ceiling would convert the high-multiplicity hang into a clean, reported failure instead of an unbounded run.
- **Pin the new boundaries in tests.** The multiplicity-explosion ($m > 8$) and construction-overflow ($10^{308/n}$) cliffs are not yet captured by the test suite; pinning them makes future fixes show up as deliberate changes.

Generated from live measurements of the Phase-1 CPU build. Tables reflect single-thread Release/-O2 runs; absolute timings are machine-dependent, but the scaling trends and break points are intrinsic to the algorithm.